# ARTICLE REVIEW TEMPLATE

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## Brief summary

This paper introduces the **RE4HCAI (Requirements Engineering for Human-Centered AI)** framework, a novel approach designed to integrate human-centered aspects into the early stages of AI software development. The authors argue that existing AI development often overemphasizes technical components while neglecting crucial human factors like biases, user needs, and ethical considerations, leading to irresponsible and non-inclusive systems.

To address this gap, the framework was developed by synthesizing industry guidelines (from Google, Microsoft, Apple), academic literature, and an expert survey. The framework is structured into six key areas: **(1) user needs, (2) model needs, (3) data needs, (4) feedback and user control, (5) explainability and trust, and (6) errors and failure.** To make the framework practical, the authors provide a requirements catalog (checklist) for elicitation and a conceptual modeling language for visual representation. The framework's utility is demonstrated through a case study involving a deep learning system designed to enhance the quality of 360° videos for virtual reality (VR) users. The authors conclude that the RE4HCAI framework successfully helps teams identify and prioritize human-centered requirements, revealing critical considerations that were initially overlooked in the project's traditional development process.

## Key research points and their importance

- **A Structured Framework for Human-Centered AI Requirements:** The core contribution is the **RE4HCAI framework itself**. It consolidates a wide range of abstract guidelines and research into a structured, actionable process with six distinct focus areas.
  - **Importance:** This is significant because it moves the conversation about "ethical" or "human-centered" AI from high-level principles to a concrete engineering practice. It gives teams a tangible starting point and a common vocabulary for discussing and documenting these critical, yet often overlooked, non-functional requirements.
- **A Practical Elicitation Catalog and Visual Model:** The framework is supported by two key artifacts: a detailed checklist-style catalog and a visual modeling language. The catalog operationalizes the six areas into specific questions and prompts for requirements engineers. The visual model helps diverse teams (including non-engineers) understand the system's human-centric goals and constraints.
  - **Importance:** These tools bridge the gap between theory and practice. A framework is only useful if it can be applied. The catalog makes the elicitation process systematic and repeatable, while the visual model enhances communication and shared understanding among stakeholders with different backgrounds (e.g., data scientists, UX designers, project managers).

- **Demonstration of an Iterative RE Process for AI:** The case study reveals that many human-centered requirements for AI systems **cannot be fully specified upfront**. For instance, data biases or model limitations only became apparent after initial model training and testing.
  - **Importance:** This finding challenges the traditional, linear approach to requirements engineering and underscores the need for an iterative lifecycle in AI projects. It highlights that RE is not a one-off phase but a continuous activity that must be revisited as the AI model evolves and interacts with data. This is a critical insight for managing AI projects effectively.

## Practical application to projects

The RE4HCAI framework offers a clear roadmap that could be directly applied to real-world projects. For example, consider an open-source project aimed at developing a **mental health chatbot**. Such a project is deeply human-centric, and failure to consider these aspects could have serious consequences.

Here is how the framework's key areas could be applied:

1. **User Needs:** Instead of just defining functional requirements like "the bot should answer questions about anxiety," the framework would force us to ask deeper questions from the catalog:
   - **Automation vs. Augmentation:** Should the bot *automate* diagnosis (high-risk), or should it *augment* a user's self-help journey by providing resources and suggesting they see a professional? The framework guides us to choose augmentation to keep the user in control.
   - **Reward Function:** How do we define a "correct" answer? Is it one that keeps the user engaged, or one that provides clinically sound, safe advice? We would define the primary goal as user safety, even if it means shorter interactions.
2. **Data Needs:** The chatbot will be trained on conversational data.
   - **Bias Identification:** Using the catalog, we would proactively check for biases in our training data. Does it underrepresent certain demographics, cultures, or types of mental health issues? For example, if the data is sourced primarily from Western forums, the bot might fail to understand or appropriately respond to culture-specific expressions of distress. We would need to implement a data collection strategy to ensure diversity.
3. **Explainability & Trust:** Users need to trust the bot.
   - **Explaining Predictions:** The framework prompts us to decide *how* to explain the bot's behavior. We would implement a feature where the user can ask, "Why did you suggest this breathing exercise?" The bot could respond, "This is a common grounding technique for moments of high anxiety, based on cognitive-behavioral therapy principles." This builds trust and transparency.
4. **Errors & Failure:** What happens when the bot fails?
   - **Mitigation Plan:** We would use the framework to design a robust error-handling protocol. If the bot detects keywords related to self-harm, its primary function should be to immediately provide crisis hotline numbers and disclaim its ability to handle such situations. This "failure" plan is a critical safety requirement.

By applying this framework, the project team would move from a purely technical mindset to a holistic, human-centered one, resulting in a safer, more effective, and more trustworthy product.

## Critical evaluation — how the paper could be improved

This paper provides a valuable and timely contribution, but there are several areas where it could be strengthened.

- **Methodology:**
  **Strengths:** The multi-pronged approach of combining an SLR, industry guidelines, and an expert survey provides a robust foundation for the framework. The use of a real-world case study is a significant strength, as it grounds the theoretical framework in practice.
  **Weaknesses & Suggestions:** The evaluation relies on a single case study with one primary ML expert. While insightful, this limits the generalizability of the findings. The study would be more convincing if the framework were applied across **multiple projects in different domains** (e.g., finance, e-commerce, healthcare) to demonstrate its versatility and identify domain-specific challenges. Furthermore, the evaluation feedback is qualitative and comes from a single expert who is also an author. A more objective evaluation could involve applying the framework with teams unfamiliar with the research and measuring its impact on project outcomes, such as the number of human-centered requirements identified or the time taken for elicitation compared to a control group.

- **Scope and Assumptions:**
  **What's Missing:** The framework successfully covers the *elicitation* and *modeling* of requirements, but it does not extend to the **validation and verification** of these requirements. How does a team test whether the "explainability" or "fairness" requirement has been met? The paper could be improved by proposing or linking to methods for testing these non-functional AI requirements. Additionally, the paper focuses on the project team's perspective. It would be beneficial to include more on how to directly **involve end-users** throughout the requirements lifecycle, not just as sources of feedback for model tuning.

- **Practicality and Reproducibility:**
  **Availability:** The paper provides the full requirements catalog in the appendix, which is excellent for reproducibility and practical application. The visual modeling notations are also clearly defined.
  **How to Improve:** The modeling language, while useful, is bespoke. This creates a learning curve for new teams. The authors could improve practicality by providing a **plugin or template for an existing, widely-used modeling tool** (e.g., a UML profile or a Visio/Lucidchart template). This would lower the barrier to adoption. They could also provide a more detailed, step-by-step tutorial on how a requirements engineer would facilitate a workshop using the catalog and modeling tools.

- **Additional Studies I Would Propose:**
  A comparative study applying RE4HCAI versus a traditional RE method (like standard user stories) to the same AI project to quantitatively measure the difference in the quantity and quality of human-centered requirements identified. A longitudinal study that follows a project using RE4HCAI from inception to deployment to assess how these early requirements impact the final product and user satisfaction. An experiment to evaluate the cognitive load and usability of the proposed conceptual modeling language compared to more standard notations like UML or GORE, especially with non-technical stakeholders.

## Additional research and references

To form this review, I primarily focused on the provided article itself. However, reflecting on its context, the following sources and concepts were influential:


1. Shneiderman, B. (2022). *Human-Centered AI*. Oxford University Press. This book provides the broader intellectual foundation for the paper's entire premise. Shneiderman is a key figure in this field, and his work on creating AI systems that are reliable, safe, and trustworthy heavily overlaps with the goals of the RE4HCAI framework. His emphasis on human control and oversight is a principle that runs through all six areas of the framework.

2. Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., ... & Inkpen, K. (2019). Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI conference on human factors in computing systems* (pp. 1-13). This work is explicitly cited by the authors as a source. Consulting it directly helps to understand the origin of many concepts in the "Feedback & Control" and "Explainability & Trust" areas. It provides a design-level perspective that complements the requirements-level focus of the RE4HCAI framework.

3. Vogelsang, A., & Borg, M. (2019). Requirements engineering for machine learning: Perspectives from data scientists. In *2019 IEEE 27th International Requirements Engineering Conference Workshops (REW)* (pp. 245-251). IEEE. This paper highlights the gap between traditional RE and the needs of ML projects from the data scientist's point of view. It corroborates the findings of the reviewed paper, particularly the idea that RE for AI must be more iterative and data-centric, and confirms that the problem the authors are trying to solve is a recognized and significant one in the field.